

# Raghav Senthil Kumar

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## EDUCATION

### University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

## EXPERIENCE

### Founding Software Engineer | Clusion

September 2025 – January 2026

- Engineered the backend document search engine in *Python (FastAPI)* for a competitive high school debate research SaaS platform hosted on *AWS*, developed by an *Agile* team of **10 engineers** and drawing **10K+ site visitors**
- Indexed **600K+ debate-evidence documents** on *Elasticsearch* via *FastAPI*, returning results in **under 1 second**
- Designed and built a text-retrieval algorithm that extracts evidence segments from debate cases by validating source URLs and filtering statistical outliers with **IQR** on the word-count distribution of tag, citation and evidence text

### Product Development Intern | Jetson

January 2025 – February 2025

- Conducted **7 iterations of UI/UX testing** across 3 experimental features on the Jetson App (100K+ users)
- Delivered a **5-page report** recommending strategies to optimize user engagement directly to Jetson's CEO
- Redesigned user onboarding on *Figma*, cutting it from **12 to 5 pages**, validated via **A/B testing** on 30 test users

### Data Analytics Intern | STEM E

June 2024 – July 2024

- Analyzed 3 years of TikTok data (**200+ posts**) with *Python (pandas)*, *Hootsuite*, *Power BI* and *Microsoft Excel*
- Identified that “science fun facts” posts comprised 69.3% of views, despite accounting for 7% of all content
- Pitched a “fun facts” strategy to the Executive Director, who adopted it; engagement rose **24% (8K+ views)**

## PROJECTS

### Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Created a dual-API *ETL* pipeline ingesting papers from *OpenAlex* & backfilling metadata from *Semantic Scholar*
- Stored paper data on *Supabase (PostgreSQL)* and wrote *SQL* queries to filter papers by field, year and citation count
- Built an interactive citation graph (*React Flow*) that maps a paper's outbound citations across **26 research fields**
- Deployed a **RAG agent (Claude Sonnet 4.5)** that retrieves session-scoped vector documents (*Supernemory*)

### Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Prompt-engineered a *JSON schema* for a **Claude Sonnet 4.6 agent** to extract condition, brand, model and year
- Revised the schema after each wrong prediction until **10 consecutive images** were correct on each of **3 features**
- Built a *quantile regression* statistical model on the extracted features, reaching **87% accuracy** on **30 samples**
- Scraped **15,000** truck-listing images for *data mining*, auditing data quality via de-duplication and non-USD filtering

### SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Built a *Python FastAPI* data pipeline that ingests live water, waste & energy data from 3 tools, runs **4 Random Forest models** for anomaly detection & delivers actionable steps via **Gemma Voice LLM** on a *Next.js* dashboard
- Engineered a **suite of 3 edge AI tools** that collects data regarding a school's energy, water and food consumption
- Led a team of 5 across project ideation, development and presentation. **Finalist at USAII Global Hackathon**

### EcoBin | Research Paper | [Source Code](#)

March 2026 – May 2026

- Designed a two-stage *EfficientNetV2-S* neural network (**4.39M** parameters) in *PyTorch*, trained on **15K+ images** to classify waste into 4 disposal pathways at **96.13% accuracy**. Published a paper on *arXiv* and 2 datasets on *Kaggle*
- Built a synthetic data generation pipeline using *U2-Net* segmentation and alpha-compositing to generate 9K images to train the model to detect contamination in recyclable waste. **Placed 2nd at the Arizona State Science Fair**
- Engineered a smart bin prototype using *Arduino Uno* that sorts the waste item with **less than 2 second latency**

## SKILLS

**Data & Analytics:** Microsoft Excel, Python, SQL, Power BI, pandas, NumPy, Hootsuite, A/B Testing, Kaggle

**Languages & Frameworks:** Java, C++, JavaScript, HTML/CSS, FastAPI, Flask, Next.js, React, React Flow

**Databases & Cloud:** Google Cloud, Elasticsearch, PostgreSQL, MySQL, Firebase, Supabase, AWS, Vercel, Docker

**AI/ML:** PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Grad-CAM, Hugging Face, Claude API, Gemma 4, RAG

**Concepts:** ETL, Prompt Engineering, LLMs, Data Quality, Data Pipelines, Data Visualization, Generative AI